

UNIVERSITY OF SCIENCE

Faculty of Information Technology

PROJECT PROPOSAL

GymHub

Smart Gym Management System

Course: Introduction to Software Engineering

Class: 24C07

Group: 01

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Date: June 22, 2026

Contents

1	Project Resources	2
1.1	Team Members	2
1.2	Equipment	2
1.3	Budget	2
1.4	Documentation Management	2
2	Introduction	2
2.1	Problem Statement	2
2.2	Proposed Solution	3
2.3	Motivation	3
3	Target Users and Environment	4
3.1	Target Users	4
3.1.1	Gym Members	4
3.1.2	Administrators	4
3.2	Operating Environment	4
4	Key Features	5
4.1	Authentication	5
4.2	Membership Management	5
4.3	Workout Plan Management	5
4.4	Progress Tracking	5
4.5	Administration	5
5	User Interviews	6
5.1	Interview Objective	6
5.2	Interview Participants & Limitations	6
5.3	Interview Findings & Feature Derivation	6
5.3.1	Gym Member Requirements	6
5.3.2	Personal Trainer Features (Descoped)	7
5.3.3	Administrative Requirements (Derived)	7
5.4	Evidence	7
6	Conclusion	10
7	References	10

1. Project Resources

1.1 Team Members

Student ID	Full Name	Role
24127073	Khả Phước Lộc	Project manager, FrontEnd Developer
24127041	Hoàng Trung Hiếu	BackEnd Developer, Database Management, Tester
24127125	Nguyễn Thanh Phát Thịnh	Frontend Developer
24127264	Trần Thái Vỹ	BackEnd Developer, Tester

1.2 Equipment

The project will be developed using the personal laptops of all team members.

1.3 Budget

Gymhub is built to mimic real software development without any monetary investment. Every service is used on its free tier, all labour is contributed by the student team, and all hardware is equipment the members already own. The **total project cost is 0 VND**.

1.4 Documentation Management

Project documents will be maintained and updated by the team member responsible for the corresponding task or deliverable.

2. Introduction

2.1 Problem Statement

The fitness industry has experienced significant growth in recent years, leading to a surge in gym memberships. However, many facilities still rely on fragmented, manual methods to manage daily operations. Membership records are often maintained via disjointed spreadsheets, workout plans are distributed through paper documents or static chat messages, and facility capacity is left untracked.

These outdated practices create distinct operational bottlenecks:

- **Gym Members** struggle to maintain workout consistency, track workout routines, monitor historical fitness progress, manage subscription renewals, or gauge real-time gym overcrowding before arriving.

- **Gym Administrators** face administrative friction in maintaining accurate user databases, processing manual billing, broadcasting announcements, and gaining clear data insights into gym performance.

Ultimately, the lack of a centralized platform results in data duplication, operational inefficiencies and a degraded user experience.

2.2 Proposed Solution

To address these challenges, we propose *GymHub*, a centralized, web-based gym management and fitness tracking platform designed for gym members and administrators.

The system streamlines gym operations by replacing fragmented workflows with a single, unified digital ecosystem. For members, *GymHub* provides the flexibility to freely customize personal workout plans or utilize automated workout generation for personalized guidance and driving accountability through an interactive workout streak and consistency system. while also visualizing progress analytics and tracking real-time gym occupancy. For administrators, it provides a dedicated dashboard to oversee memberships, automate billing, and analyze facility utilization.

2.3 Motivation

The development of *GymHub* is driven by the following objectives:

- **Enhance Operational Efficiency:** Transition gyms from manual spreadsheet tracking to automated administrative workflows.
- **Empower Member Autonomy:** Provide users with smart, self-guided tools for progress tracking and personalized workout generation without requiring constant staff intervention.
- **Foster Long-Term Retention:** Leverage behavioral design patterns, such as workout streaks and consistency metrics, to keep gym members engaged and reduce the drop-out rate.
- **Optimize Facility Management:** Mitigate peak-hour overcrowding through data-driven occupancy monitoring.
- **Academic Integration:** Apply modern software engineering methodologies, database design principles, and full-stack development practices to a real-world problem.

3. Target Users and Environment

3.1 Target Users

3.1.1 Gym Members

Main goals:

- **Design and build** customized workout plans with optional automated support.
- **Access and manage** gym membership information and digital profiles.
- **Track and visualize** physical fitness progress and historical performance metrics over time.
- **Maintain active workout streaks** and consistency metrics to drive long-term motivation and habit-building.
- **Receive real-time notifications** for scheduled workout reminders and live gym crowding alerts.
- **Process online payments** securely for membership renewals and plan upgrades.

3.1.2 Administrators

Main goals:

- **Centralize membership management** by tracking subscription statuses, processing renewals, and maintaining accurate member profiles.
- **Monitor business financial performance** through integrated revenue reporting, payment tracking, and basic operational analytics.
- **Configure facility capacity thresholds** and broadcast gym-wide notifications or critical system alerts to members.

3.2 Operating Environment

- Platform: Web Application
- Client: Desktop and Mobile Browsers
- Internet Connection Required

4. Key Features

4.1 Authentication

- User registration
- Login and logout
- Password recovery

4.2 Membership Management

- Membership renewal
- Membership status tracking
- Admin approves or rejects membership renewal request

4.3 Workout Plan Management

- Members design customized workout plans with LLM support.
- Members view assigned workout plans.
- Workout completion tracking.

4.4 Progress Tracking

- Weight tracking
- Body measurement tracking
- Progress history

4.5 Administration

- User management
- Membership management
- Dashboard and statistics
- Broadcast gym-wide notifications

5. User Interviews

5.1 Interview Objective

Using Google Forms and direct discussions, this study aims to identify the specific needs, pain points, and desired features of key stakeholders regarding gym management and fitness tracking systems.

5.2 Interview Participants & Limitations

The initial research design targeted three primary stakeholder groups: Gym Members, Personal Trainers (PTs), and Gym Administrators.

Methodology Limitation: Due to scheduling constraints and limited availability, administrative personnel could not be reached during the data collection window. Consequently, primary data reflects only the perspectives of gym users and PTs. To mitigate this gap, administrative requirements were derived from secondary research, competitive analysis of existing gym management software, and industry best practices.

5.3 Interview Findings & Feature Derivation

Analysis of the participant responses revealed several recurring challenges. These insights directly shaped the functional scope of the proposed system, *GymHub*.

5.3.1 Gym Member Requirements

- **Smart Workout Plan Generation:** Respondents reported widespread uncertainty regarding exercise selection. To address this, the system will feature an automated generator that recommends personalized workout routines based on user goals, fitness levels, and training preferences.
- **Fitness Progress Tracking:** Difficulty monitoring physical progress was a primary pain point. The system will provide dedicated tools to record and visualize fitness journeys through historical logs, statistics, and charts.
- **Workout Streak and Consistency Tracking:** Interview data revealed that many users frequently skip sessions due to a lack of immediate accountability. *GymHub* will implement a dynamic Workout Streak system. By tracking consecutive active weeks or scheduled days completed, creating a strong psychological incentive for users to maintain their active streaks and build long-term habits.

- **Real-Time Notifications & Occupancy Monitoring:** To improve workout consistency, the system will send automated schedule reminders. Furthermore, to combat the frequent frustration of peak-hour overcrowding, real-time gym occupancy monitoring will be integrated to help users plan their visits efficiently.
- **Digital Membership & Payment Management:** To streamline administrative friction, the platform will support online membership renewals, digital payment processing, and instant access to membership status.

5.3.2 Personal Trainer Features (Descoped)

Interview analysis revealed low demand for personal trainer booking functionalities, as most respondents prioritized self-guided workout tools over direct trainer interaction. Given these findings and the strict development timeline, the **Personal Trainer module has been removed from the project scope** to allow deeper focus on user-centric automated features.

5.3.3 Administrative Requirements (Derived)

Based on secondary research and system dependencies, the following core administrative features are integrated to support the platform's ecosystem:

- **Centralized Membership Oversight:** Tools for administrators to manage member records, track subscription statuses, process manual renewals, and maintain accurate user databases.
- **Financial Reporting & Analytics:** A dashboard offering visibility into payment records, membership revenue, and key business performance metrics to support data-driven operational decisions.

5.4 Evidence

The primary research data collected from our target audience can be reviewed directly via the following link: [Google Forms Interview Responses](#).

The chart data highlighting participant behaviors, pain points, and feature preferences are presented below:

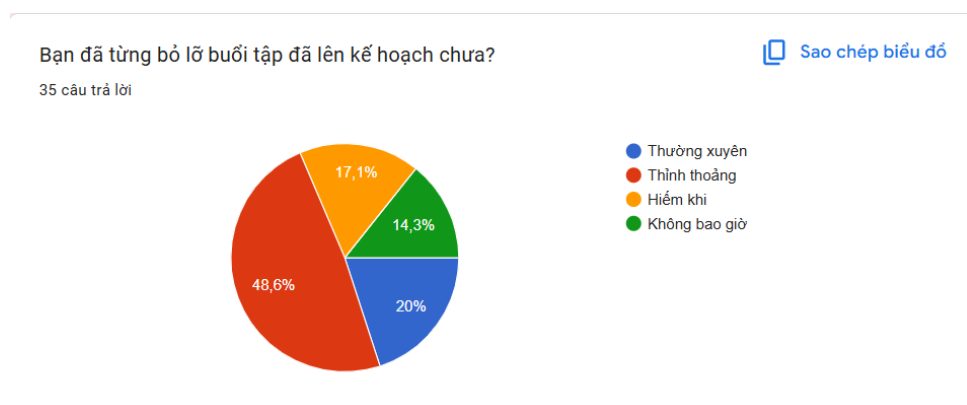


Figure 1: Gym members frequently skip sessions

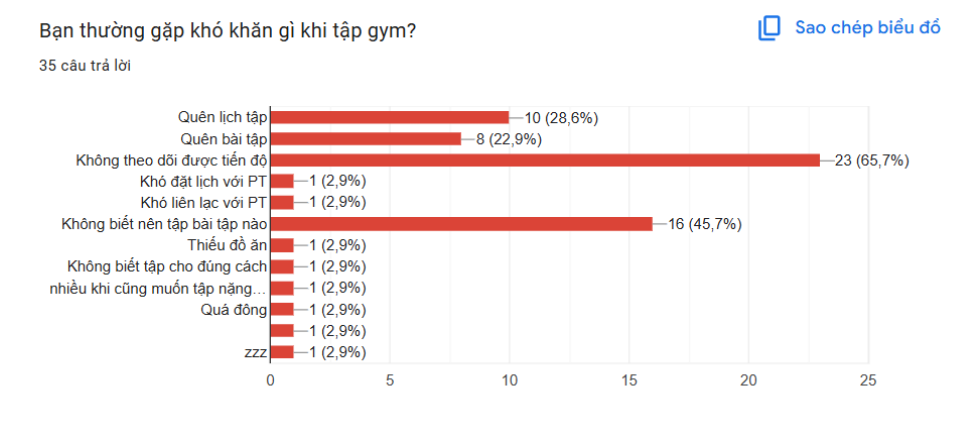


Figure 2: Gym member's difficulties

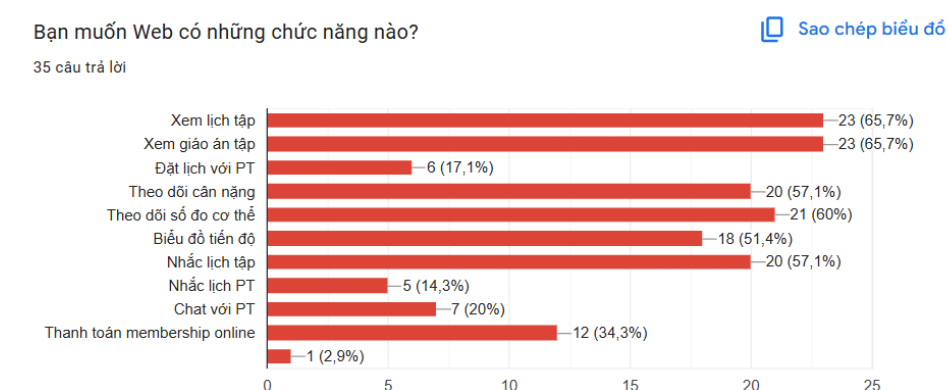


Figure 3: Gym member's desired features

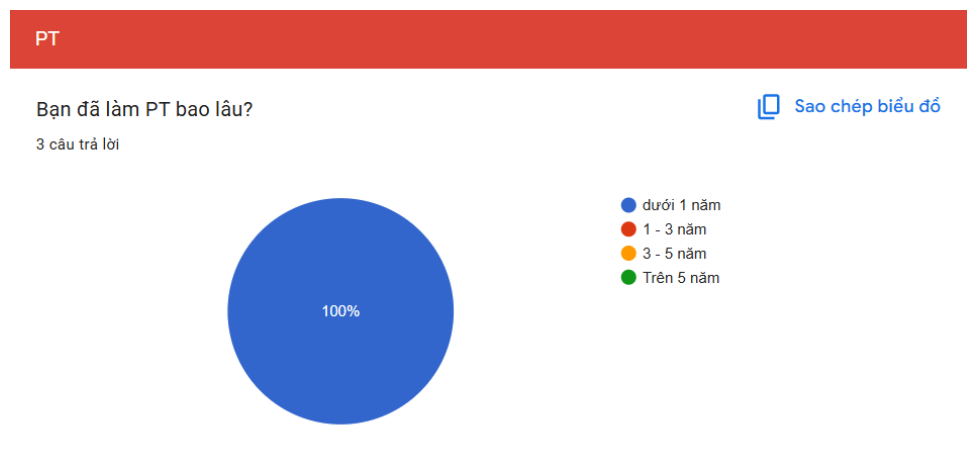


Figure 4: A small number of candidates were interviewed for the PT position.

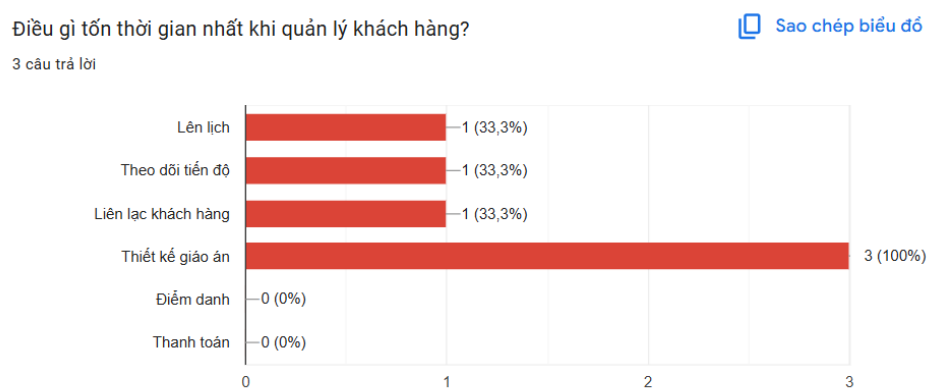


Figure 5: Personal trainer's difficulties

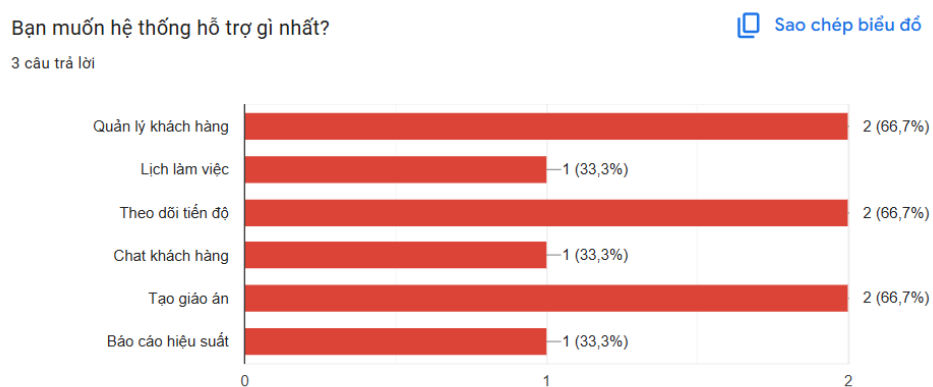


Figure 6: Personal trainer's desired features

6. Conclusion

In conclusion, the modern fitness industry requires digital solutions that move beyond fragmented spreadsheets and paper-based tracking. *GymHub* addresses this need by providing a streamlined, web-based platform that unifies member self-management and administrative operations into a single ecosystem. By offering members the autonomy to customize workout routines or generate automated plans, track their fitness metrics, and monitor real-time gym occupancy, the system directly tackles the core frustrations of overcrowding and administrative friction.

Simultaneously, the platform equips administrators with the tools necessary for efficient membership oversight and data-driven financial reporting. Ultimately, this project not only solves a practical, real-world operational challenge within gym management but also serves as a robust framework for applying advanced software engineering methodologies, full-stack development practices, and database management principles in an academic environment.

7. References

1. [finegym](#)
2. [fitxzo](#)
3. [GymOpsHQ](#)
4. [Interview result \(Google form\)](#)